

Robust Gesture Recognition with mmWave Sensing using Supervised Contrastive Learning

Kai-Siang Wang
kw37@illinois.edu
University of Illinois Urbana
Champaign
Urbana, IL, USA

Jiayi Xiao
jiayix8@illinois.edu
University of Illinois Urbana
Champaign
Urbana, IL, USA

Hongyi Jiang
hongyi9@illinois.edu
University of Illinois Urbana
Champaign
Urbana, IL, USA

CCS CONCEPTS

• **Human-centered computing** → Interaction techniques; • **Computing methodologies** → Neural networks; Supervised contrastive learning; Data augmentation.

KEYWORDS

Datasets, Neural networks, Gesture recognition

ACM Reference Format:

Kai-Siang Wang, Jiayi Xiao, and Hongyi Jiang. 2025. Robust Gesture Recognition with mmWave Sensing using Supervised Contrastive Learning. In *Proceedings of Smart-X '22: Fake ACM Symposium on Smart cities, homes, phones, and beyond (Smart-X '22)*. ACM, New York, NY, USA, 7 pages. <https://doi.org/10.1145/nnnnnnn.nnnnnnn>

ABSTRACT

Human gesture recognition is essential for intuitive and contactless human-computer interaction while preserving user privacy and operating well across different lighting conditions. However, radar-based gesture recognition faces two major challenges: the scarcity of large, high-quality public datasets and the significant performance degradation caused by environmental variations, especially in long-range, and complex environment with multipath interference. In this paper, we propose a gesture recognition framework that combines supervised contrastive learning (SupCon) with physically-motivated radar data augmentation. We evaluate our approach using the open-source DI-Gesture dataset and additional real-world data collected at varying distances. Experimental results demonstrate that our method consistently improves recognition robustness and outperforms the baseline by up to 3.36% across cross-environment and cross-location settings, highlighting its potential for reliable deployment in diverse real-world scenarios.

1 INTRODUCTION

Human gesture recognition plays an important role for human-computer interaction, as it provides an intuitive interface for humans to remotely control the devices. While common approaches such as camera-based vision [7] and wearable devices [5] have been widely adopted, each comes with limitations. Cameras are highly susceptible to lighting conditions and physical obstacles, and they also raise privacy concerns due to captured images [1]. Wearable devices, on the other hand, are often considered inconvenient because they require continuous physical contact and frequent charging.

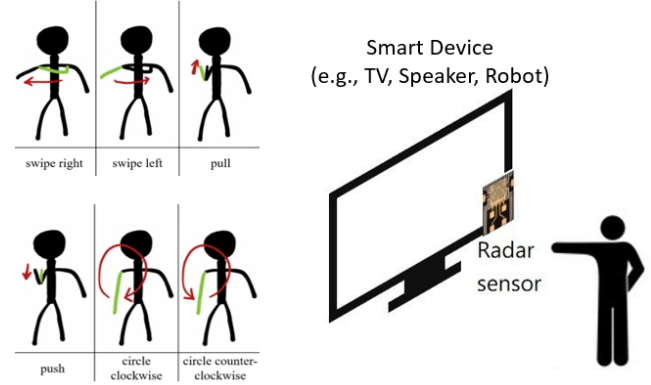


Figure 1: Overall Concept of Our Proposed Approach

Radar sensing, as an emerging compelling alternative, does not require lighting or physical contact and collects abstract object data rather than personal visual information. The current state-of-the-art gesture recognition systems leverage these advantages by training deep-learning models on radar signals, typically represented as point clouds or radar images containing information like angle, velocity, and range [2, 4, 6].

However, radar sensing also faces several limitations. Unlike the field of computer vision or natural language processing, radar sensing does not have extensive, high quality, and public gesture datasets [4]. Researchers often rely solely on self-collected data acquired under limited experimental conditions, which hinders the development of models with robust generalization capacities. To mitigate this data scarcity, various data augmentation methods have been proposed to diversify training datasets. For example, by adding random noise or extrapolating through rotation and shifting of signal images based on a real dataset, more training datasets can be generated. While these approaches can increase dataset size, they often provide limited improvements in robustness, particularly when the model is deployed under new sensing conditions.

Another critical challenge arises from environmental variations, particularly in long-distance scenarios (e.g., beyond 5 m) where radar returns are weak and highly susceptible to multipath effects caused by reflections from surrounding objects. These conditions can distort radar features and significantly decrease recognition accuracy in real-world deployments.

To address these challenges, this work explores the use of supervised contrastive learning (SupCon) [3] in combination with

physically motivated data augmentation to improve the generalizability of mmWave radar gesture recognition models. SupCon extends traditional contrastive methods by using class labels to create more positive and negative pairs to learn more discriminative representations. Unlike unsupervised contrastive learning that only compares two augmented views from the same sample, the supervised approach treats all samples from the same gesture class as positive pairs while treating different classes as negatives. Then, by bringing similar samples (positive pairs) closer and pushing dissimilar ones (negative pairs) apart in the embedding space, SupCon helps the model to generalize on cross-domain variations (e.g., different users, positions, environments), while still keeping clear separation among different gesture classes (e.g., clockwise, slide right, etc).

On top of the supervised contrastive learning framework, we proposed two physics-based data augmentations that simulate realistic radar propagation effects. Radar measurements often vary with distance, multipath reflections, and background noise, which can lead to unstable feature embeddings. Unlike conventional data augmentation methods that apply general geometric or noise transformations, physically motivated data augmentation uses the physical properties of radar propagation to generate realistic signal variations. By modeling situations such as signal attenuation and multipath reflection, our augmentations help the model recognize gestures robustly across different distances and environments.

To evaluate our proposed solution, we first implemented the baseline [4] and trained the model with the baseline data augmentation techniques they proposed. Then we performed our data augmentation techniques and our supervised contrastive learning model using the open-source datasets provided by DI-Gesture[4]. Finally, we collected new data from various distances to compare the performance of our model with the baseline model.

We summarize the main contributions of this work as follows:

- (1) **Contrastive learning framework for radar sensing:** We propose a supervised contrastive learning-based training framework for mmWave radar gesture recognition, enabling the model to learn distinct representations across diverse sensing conditions.
- (2) **Physically inspired data augmentation:** We introduce two new augmentation techniques designed to emulate realistic radar signal variations:
 - *Distance-aware attenuation:* simulates far-range sensing by attenuating signal strength to mimic weaker returns at longer distances.
 - *Multipath emulation:* simulates environmental reflections by modifying the original processed radar image through controlled offsets, rotation, and signal reduction, then reinserting the transformed reflections into the radar data to reproduce multipath effects.
- (3) **Comprehensive evaluation:** We compared in-domain and cross-domain classification results using both the baseline approach and our proposed approach, demonstrating improvement on all scenarios with our proposed method.

Approach	Noise	Shifting	Attenuation	Multipath
Rodar [2]	✗	✗	✗	✗
GesturePrint [6]	✓	✗	✗	✗
DI-Gesture [4]	✓	✓	✗	✗
Ours	✓	✓	✓	✓

Table 1: Elements Considered in Data Augmentation of Each Models

2 PRIOR SOLUTIONS

Existing research in mmWave radar gesture recognition has explored diverse methodologies. The typical workflow starts with collecting and labeling gesture radar signals, applying basic noise filtering, and training a neural network to classify the gesture radar image sequence. The current approaches primarily differ in dataset characteristics, data augmentation techniques, and neural network structures.

Data Augmentation. Due to the lack of large, public datasets for mmWave radar gestures, most prior works rely on self-collected datasets, which are limited in terms of distance, angle, and environment. Furthermore, they often only apply simple data augmentation techniques like Gaussian Noise. So the training datasets lack diversity, making the models vulnerable to changes in setting. DI-Gesture [4] addresses the data bottleneck by proposing a more sophisticated data augmentation framework to generate samples by shifting the radar image in angles, speeding or reversing the radar image sequence, and rotating radar images. It successfully improves the robustness of the classifier for situations not included in the original datasets. However, these augmentation techniques didn't fully consider the effect of the surrounding environment, like multipath effect. It also ignored the signal attenuation when applying shifting to the radar image, therefore making it less robust to recognize gestures at further distances.

Neural Network-based Approach. Prior works primarily address the high-dimensional radar signals by employing neural networks for gesture classification. Some studies utilize specialized architecture for specific challenges in recognition. For example, Rodar [2] divides a single radar signal image into small patches, and use self-attention mechanism to distinguish gesture signal from background activities. GesturePrint [6] not only classify gesture but also recognize user based on the same point cloud image. Despite these architectural advancements, both system exhibit a noticeable decline in accuracy when recognizing gestures at greater distances(>5m). They also lack evaluations under new or complicated environments. This shared limitations show a substantial remaining opportunity to improve both robustness and generalizability in mmWave radar gesture recognition tasks.

Different from existing studies, our proposed system includes two physically-motivated data augmentation techniques that simulate real-world distance attenuation and multipath effect. In contrast to the prior works, we use a supervised contrastive learning model to learn more discriminative gesture representations from these augmented samples, improving generalization across different users, environments and distances.

3 SYSTEM DESIGN

To address the above limitations, we propose a framework that improves gesture recognition robustness through two key components: (1) physically-motivated augmentations that simulate signal attenuation and multipath effects, and (2) supervised contrastive learning that learns discriminative representations from augmented samples.

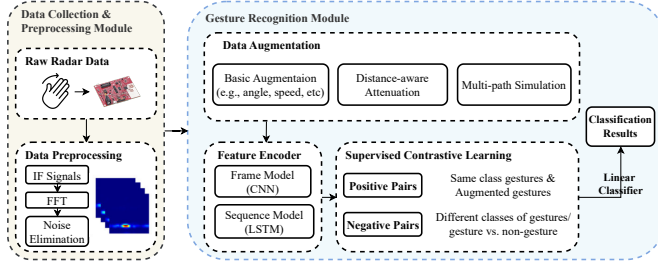


Figure 2: System Overview

3.1 Data Collection & Preprocessing

In this module, raw radar signals are acquired using the AWR1843 and DCA1000EVM devices. The collected signals are transformed using the Fast Fourier Transform (FFT) to obtain Range-Doppler-Angle (RDA) representations. To further reduce the noise, Doppler summation is applied to integrate energy across Doppler bins, and a Doppler frequency threshold is used to remove static background objects. The resulting processed radar data forms sequential Dynamic Range-Angle Images (DRAI), which is passed to the Gesture Recognition module.

3.2 Data Augmentation

To enhance the diversity and robustness of the radar dataset, data augmentation techniques are applied during gesture recognition. In addition to the basic augmentations from the baseline work, such as spatial shifting (angle and range translation) and temporal scaling (speed variation), we propose two physically inspired data augmentation techniques: *Distance-Aware Attenuation* to simulate signal weakening at longer sensing distances, and *Multipath Emulation* to mimic reflections from surrounding environments.

Distance-Aware Attenuation. The received signal strength is inversely proportional to the distance between the object and the antenna. This relationship is shown in the equation below:

$$\frac{P_r}{P_t} = \left(\frac{\lambda}{4\pi d} \right)^2 \quad (1)$$

where P_r is the received power, P_t is the transmitted power λ is the wave length, and d is the distance between the object and antenna. To simulate this effect of shifting for a random distance, we reduce the power strength in the raw data proportional to difference of square of original distance and the shifted distance. Then we shift the whole radar image in the range coordinate (Y-axis). As shown in Figure 3, the main characteristics of the image is still preserved, while the power is reduced and the image is shifted further away.

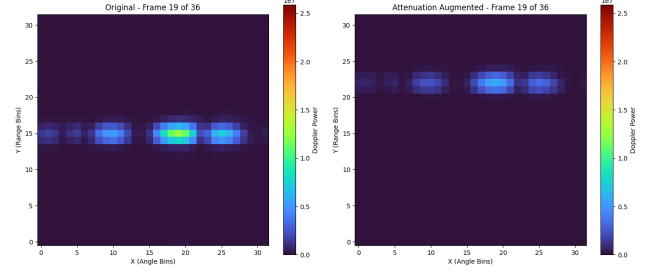


Figure 3: Example of Distance Attenuation

While this is a simplified model that only considers range-dependent attenuation, without considering angular effects at greater distances, it provides a computationally efficient simulation that captures the primary distance-dependent signal variation.

Multipath Emulation. We emulate multipath effect by producing a mirrored version of the radar image with a shift in both angle and range coordinates and overlapping it with the original image. This simulates the effect of a nearby flat metallic surface producing a reflect of the gesture signal at a slightly further place on the side. An example is shown in Figure 4. There is a dim shadow of the gesture at the top right corner of the augmented image.

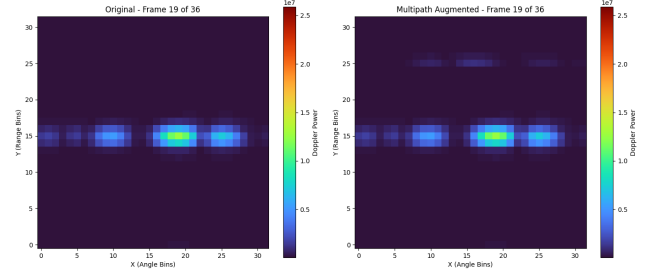


Figure 4: Example of Multipath Effect

Even though this is a relatively naive implementation of the multipath effect, as the multipath-rich environment would usually have much more complicated result, our simulation effectively models the noises that remain correlated to the original image.

3.3 Supervised Contrastive Learning

Our training follows a two-stage approach. In **stage 1 (super contrastive learning)**, we randomly apply data augmentation on each radar frame sequence to create two augmented views of the same sample. Both views are processed by the feature encoder, which consists of a convolutional frame model (CNN) to extract spatial features and a two-layer LSTM to capture temporal features across frames. The encoder outputs are then passed through a projection head to produce normalized embeddings. The supervised contrastive loss is computed over these embeddings.

Specifically, for a batch of B samples, each sample is randomly augmented twice, producing $2B$ embeddings. Positive pairs are defined as: (1) the two augmented views of the same sample, and (2) samples from the same gesture class. All other sample pairs

are treated as negative pairs. The supervised contrastive loss [3] is designed to pull positive pairs closer while pushing negative pairs apart. For example, when processing a "clockwise" gesture sample, the loss ensures its two augmented views are similar, brings it closer to other "clockwise" samples, and separates it from different gestures like "push". This enables the encoder to learn features that are both robust to data augmentations and discriminative across gesture classes.

For each anchor embedding z_i , the loss [3] is:

$$\mathcal{L} = - \sum_{i \in I} \frac{1}{|P(i)|} \sum_{p \in P(i)} \log \frac{\exp(z_i \cdot z_p / \tau)}{\sum_{a \in A(i)} \exp(z_i \cdot z_a / \tau)} \quad (2)$$

where $P(i)$ contains all positive samples for anchor i , $A(i)$ contains all samples except i , and τ is the temperature parameter to control the sharpness of the similarity distribution.

In **stage 2 (linear classification)**, we freeze the trained encoder (CNN + LSTM) and train a linear classifier on top of the extracted features using cross-entropy loss to perform gesture classification. This two-stage approach allows us to evaluate the quality of learned features independently. By using a simple linear classifier in stage 2, the model's performance directly reflects the high quality of the learned representations rather than the complexity of the classifier.

4 IMPLEMENTATION

4.1 Baseline Reproduction

The baseline DI-Gesture dataset consists of 24050 samples from 25 volunteers, 5 locations and 6 environments. It includes six predefined gestures: Push (PH), Pull (PL), Left Swipe (LS), Right Swipe (RS), Clockwise Turning (CT), and Anticlockwise Turning (AT). In addition to these gestures, a set of negative samples is provided to represent non-gesture activities. There are 5 locations in the dataset, which are (0.6m, 0°), (0.8m, 0°), (1.0m, 0°), (0.8m, -30°) and (0.8m, 30°). Each sample of gesture was processed into a sequence of frame, which consist of a 2D array of 32 x 32 representing the signal magnitude in range and angle. The time period for each frame is 50ms, and the total length of the gesture is determined by the gesture segmentation algorithm. Therefore, the number of frame for gesture ranges from 9 to 98, with median of 28.

We obtained the open-sourced dataset and code from DI-Gesture. We followed the training steps described in the paper and integrate the data augmentation techniques mentioned in the paper into the training process. The model was trained with 200 epochs. Further comparison with our proposed approach is made in the Evaluation section.

4.2 Data Collection

To evaluate the performance of the baseline model and our model under new environment, we collected gesture data from 3 volunteers, 2 environments, and 5 locations. Fig. 5 shows different setups we used for data collection. The environments are mainly different in the furniture and wall placements, which results in different multipath effects. We collected the same six gestures performed in DI-Gesture's dataset. The locations of the user are (1m, 0°), (2m, 0°), and (3m, 0°). Each user performed each gesture 5 times with the guidance of the LED light on the board. In total, we have collected

180 samples. The details are shown in Table 2.

4.3 Device Configuration

We collected the raw radar data using the TI AWR1843 radar and DCA1000 real-time data acquisition board, the same devices used in DI-Gesture. Each frame has 128 chirps and each chirp has 128 sample points. This is the same setting listed in DI-gestures. However, the format of raw data in the open-source dataset doesn't always follow this setting. We aligned with the setting from the paper for consistency. The radar collects 20 frames per second, and each collection last for 2 seconds. We activated 4 receiving antennas and 2 of 3 transmitting antennas to match the configuration of DI-Gesture. We used the data processing program from DI-Gesture to transform raw data into DRAI, a sequence of frame with resolution of 32 x 32 bins in range and angle, the same format as the baseline dataset.

4.4 Supervised Contrastive Learning Model

Our SupCon model follows the same encoder architecture as DI-Gesture [4]. The feature encoder begins with three convolutional layers with 16, 32, and 64 filters (kernel size 3 x 3, stride 2). Each convolutional layer is followed by batch normalization and ReLU activation. The stride of 2 in each layer progressively reduces spatial dimensions from 32 x 32 to 3 x 3. The resulting feature map is flattened (576 dimensions) and fed into a fully connected layer with dropout rate 0.5, producing a 512-dimensional feature vector for each frame. These per-frame features are then fed into a two-layer LSTM with hidden size 256 to capture temporal features across frames. The CNN and LSTM together form the feature encoder, producing a 256-dimensional representation.

For supervised contrastive learning, the encoder output is passed through a two-layer projection head with ReLU activation and 0.5 dropout to produce a 64-dimensional embedding, which is L2-normalized before computing the contrastive loss. During linear evaluation (stage 2), the projection head is not used, and the 256-dimensional encoder features are directly fed into a single linear layer for 7-class gesture classification.

4.5 Training Configuration

In stage 1 (supervised contrastive learning), the encoder (CNN + LSTM) and projection head are jointly trained for 200 epochs with a batch size of 128. We used the Adam optimizer with an initial learning rate of 0.001 and weight decay of 1×10^{-4} . In stage 1, the encoder and projection head are jointly trained for 200 epochs with batch size 128. We also used a cosine annealing learning rate schedule that reduces the learning rate from 0.001 to 1×10^{-5} over the 200 epochs. The temperature parameter τ in the supervised contrastive loss is set to 0.07. In stage 2 (linear classification), we freeze all encoder parameters and trained only the linear classifier for 10 epochs using cross-entropy loss with the same optimizer and initial learning rate. Training converges within 10 epochs as only a single linear layer needs to be optimized.



Figure 5: Examples of Data Collection

Predefined Gestures	Push (PH), Pull (PL), Left swipe (LS), Right swipe (RS), Clockwise turning (CT), Anticlockwise turning (AT)
Positions	1m, 2m, and 3m away from radar
User 1	$5 \text{ samples} \times 6 \text{ gestures} \times 1 \text{ environ} \times 1 \text{ loc} = 30$
User 2	$5 \text{ samples} \times 6 \text{ gestures} \times (1 \text{ environ} \times 3 \text{ loc} + 1 \text{ environ} \times 1 \text{ loc}) = 120$
User 3	$5 \text{ samples} \times 6 \text{ gestures} \times 1 \text{ environ} \times 1 \text{ loc} = 30$
Total	180 Samples

Table 2: Detailed Description of Our Collected Dataset

5 EVALUATION

5.1 In-domain Recognition

We first evaluated the performance of our proposed solution in an in-domain setting. We trained both baseline model and our model with 80% of the data and evaluate the accuracy on the rest 20%. The confusion matrix in Figure 6 shows per-class recall rates. Both methods achieve comparable high accuracy with seen environments, users, and locations. The reason is that even though we use only 80% of the data to train the models, the size of the training dataset is still sufficiently large (19600 samples). In addition, since all data is collected in controlled environments, both models are able to effectively learn the underlying gesture patterns very well.

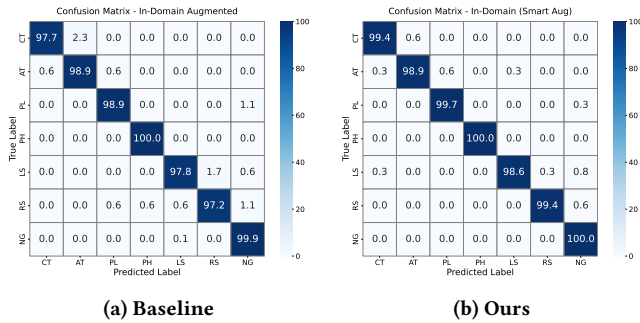


Figure 6: Confusion Matrix of In-domain Test

5.2 Cross-domain Recognition

We also compared the performance of our proposed solution with baseline method in cross-use, cross-environment and cross-location settings.

Cross-location. We evaluated cross-location generalization using leave-one-location-out, where each location is held out as the test set while training on the remaining four locations. Our method achieves an overall improvement of 1.03%, specifically better performance on location 1 (0.6m, +3.06%). This is because that baseline augmentation only applies geometric translations (vertical and horizontal shifts) of the DRAI images without considering the physical relationship between distance and signal strength. In the leave-one-location-out experiment, when testing on location 1 (0.6m), the baseline model is trained on location 2-5 (0.8-1.0m). The basic augmentation creates variations in geometric position while maintains signal strength from the original training distances, which makes it unable to generalize to the stronger signals at closer range. In contrast, our proposed augmentation shifts the DRAI in the range dimension while also applying signal attenuation based on the inverse square law of electromagnetic wave propagation. Therefore, our model can learn that a gesture performed at 0.6m with strong signal strength and the same gesture at 1.0m with weaker strength have the same pattern.

Cross-environment. We evaluated cross-environment generalization using a leave-one-environment-out, training on five environments and testing on the held-out environment. Our method achieves an overall improvement of 0.61%, where environment 6 (laboratory) shows the largest improvement of 1.71%. The six environments include meeting room, living room, bedroom, laboratory, and two office rooms, each with different multipath characteristics. The baseline augmentation lacks multipath simulation, so when trained on environment 1-5 (residential/office environments), it learns simpler multipath patterns of these layouts. However, when tested on the laboratory environment containing highly reflective objects such as metal equipment, the model performance is easily influenced by stronger reflections and multipath interference. In contrast, our proposed multipath augmentation simulates diverse

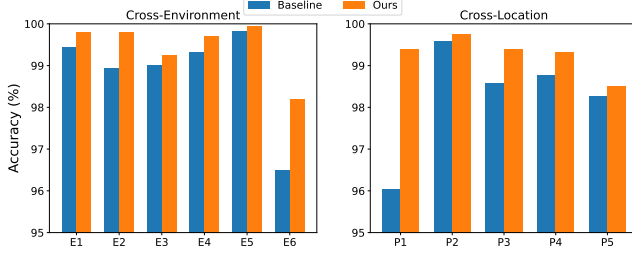


Figure 7: Cross-domain Accuracy

reflection patterns by adding mirrored, scaled-down, and attenuated copies of the DRAI at random offsets. This enables the model to learn robust representations that generalize to complex laboratory environments.

Cross-users. To evaluate generalization across users, we used the first 7 users (12250 samples) as the training data and evaluate both models with the rest of the data (users 8-25, 11800 samples). Our method achieves an overall improvement of 0.55%. The cross-user scenario is challenging because different users have variations in speed, gesture size, and etc. Unlike cross-location and cross-environment scenarios where our proposed augmentations directly address distance and multipath variations, cross-user generalization primarily benefits from the supervised contrastive learning framework combined with baseline augmentations (speed and geometric transformations) that simulate different motion variations.

The confusion matrix in Figure 8 shows some improvements on Clockwise Turning (CT) recognition and Right Swipe (RS), while some show slight decreases (e.g., Anticlockwise turning). The overall modest improvement suggests that user variability is partially captured by our proposed methods, though cross-user generalization remains a challenging problem.

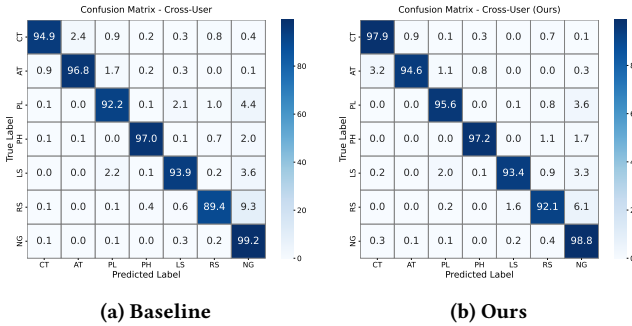


Figure 8: Confusion Matrix of Cross-User Test

Table 3 summarizes the performance across all evaluation scenarios. Our method achieves consistent improvements ranging from 0.50% (in-domain) to 1.03% (cross-location), with an overall average improvement of 0.67%. The greater improvements can be observed on cross-location (1.03%, particularly location 1 at 3.36%) and cross-environment (0.61%, particularly laboratory at 1.71%), where our proposed augmentations simulate physical effects (distance-aware

Evaluation	Baseline	Ours	Improvement
In-Domain	99.21%	99.71%	+0.50%
Cross-User	96.82%	97.37%	+0.55%
Cross-Loc p1	96.04%	99.40%	+3.36%
Cross-Loc p2	99.59%	99.76%	+0.17%
Cross-Loc p3	98.57%	99.40%	+0.83%
Cross-Loc p4	98.77%	99.32%	+0.55%
Cross-Loc p5	98.26%	98.50%	+0.24%
Cross-Loc avg	98.25%	99.28%	+1.03%
Cross-Env e1	99.43%	99.80%	+0.37%
Cross-Env e2	98.93%	99.79%	+0.86%
Cross-Env e3	99.01%	99.25%	+0.24%
Cross-Env e4	99.31%	99.69%	+0.38%
Cross-Env e5	99.82%	99.93%	+0.11%
Cross-Env e6	96.49%	98.20%	+1.71%
Cross-Env avg	98.83%	99.44%	+0.61%
Average	98.28%	98.95%	+0.67%

Table 3: Performance Summary Across All Scenarios (Accuracy)

attenuation and multipath emulation). Cross-user (0.55%) and in-domain (0.50%) show less improvements, primarily driven by the contrastive learning framework rather than our proposed augmentations, as our augmentations focus on physical variations rather than user-specific motion patterns. Overall, our approach achieves consistent improvements across all scenarios without adding inference complexity, demonstrating effective generalization to diverse real-world conditions.

5.3 Impact of Proposed Data Augmentation Techniques

We further examine how our proposed data augmentation methods affect model performance. Using our proposed two augmentations, we modify the original dataset to generate a new set, D_{aug} . We then use D_{aug} to evaluate the accuracy of the trained models for cross-user evaluation as in Section 5.2. The purpose is to see if the baseline model is robust to the attenuation and multipath effect. The results are shown in Figure 9. We can clearly see that the accuracy of the baseline model drops by 10% approximately. Meanwhile, our solution is more robust to the new dataset.

5.4 Classification of Newly Collected Data

We deployed the baseline and our model trained on the DI-Gesture's dataset to classify our newly collected gesture data. As the result shown in Figure 10, both models failed to recognize gestures, achieving total accuracy of only 0.56% and 1.13%. Specifically, both models overwhelmingly recognized valid gestures as non-gestures. The low accuracy suggests that the features of the collected data is significantly different from the training dataset. This is mostly likely due to difference in device configurations that weren't disclosed by baseline. Also, the filter setting in data processing program is based on the data collection specifically for the environments in the baseline dataset, which may not be suitable for our environments. These

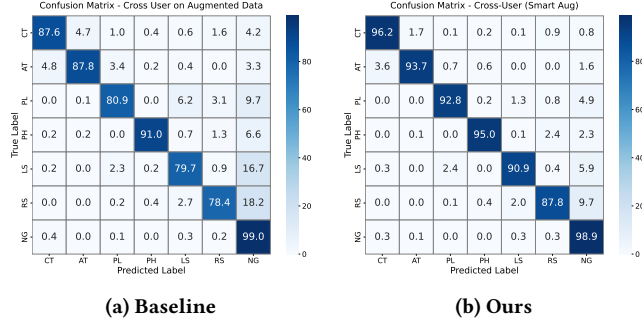


Figure 9: The Impact of Our Augmentation Techniques on the Accuracy

factors combined lead to the drastic difference between baseline dataset and our dataset.

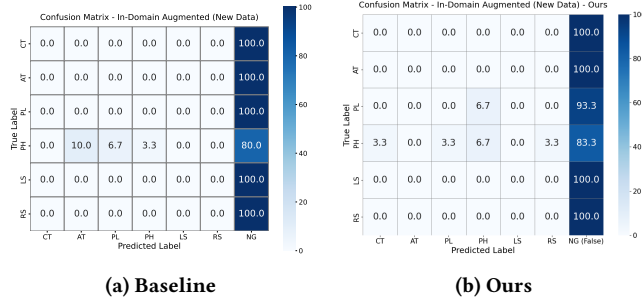


Figure 10: Classification Results on Our collected Data

6 CONCLUSION

In this paper, we presented two novel data augmentation techniques to diversify radar signal dataset and supervised contrastive learning model (SupCon), to classify gesture through mmWave radar signal images. SupCon outperformed state-of-the-art baseline model DI-Gesture in both in-domain and cross-domain settings. We also collected 180 samples of gestures from 3 users, 2 environments, and 3 positions. Our framework serves as a proof of concept that physically inspired data augmentation techniques can introduce meaningful diversity to the training dataset, and SupCon can effectively enhance robustness and generalizability for gesture recognition. Ultimately, our work offers a feasible solution to the scarcity of large public radar dataset.

Currently, performance improvement is limited to the open-source dataset. Cross-validation with our newly collected data was hindered by configuration mismatches, as the specific device configuration for DI-Gesture were undisclosed. Consequently, the effect of our data augmentation remains to be fully verified. For future work, we will first redo the data pre-processing pipeline to confirm the correctness of the processed data. We also plan to collect extensive data to allows training purely based on our data. Also, by testing under multipath-rich environment and different distances, we will be able to evaluate the robustness of our model and the

performance of data augmentation. For better data collection, we will use dynamic window segmentation to accurately targets the frames of gestures, instead of the currently fixed frame number.

REFERENCES

- [1] Karan Ahuja, Yue Jiang, Mayank Goel, and Chris Harrison. Vid2doppler: Synthesizing doppler radar data from videos for training privacy-preserving activity recognition. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*, CHI '21, New York, NY, USA, 2021. Association for Computing Machinery. ISBN 9781450380966. doi: 10.1145/3411764.3445138. URL <https://doi.org/10.1145/3411764.3445138>.
- [2] Can Jin, Xiangzhu Meng, Xuanheng Li, Jie Wang, Miao Pan, and Yuguang Fang. Rodar: Robust gesture recognition based on mmwave radar under human activity interference. *IEEE Transactions on Mobile Computing*, 23(12):11735–11749, 2024. doi: 10.1109/TMC.2024.3402356.
- [3] Prannay Khosla, Piotr Teterwak, Chen Wang, Aaron Sarna, Yonglong Tian, Phillip Isola, Aaron Maschinot, Ce Liu, and Dilip Krishnan. Supervised contrastive learning, 2020. URL <https://arxiv.org/abs/2004.11362>.
- [4] Yadong Li, Dongheng Zhang, Jinbo Chen, Jinwei Wan, Dong Zhang, Yang Hu, Qibin Sun, and Yan Chen. Towards domain-independent and real-time gesture recognition using mmwave signal. *IEEE Transactions on Mobile Computing*, page 1–15, 2022. ISSN 2161-9875. doi: 10.1109/tmc.2022.3207570. URL <http://dx.doi.org/10.1109/TMC.2022.3207570>.
- [5] Sheng Shen, He Wang, and Romit Roy Choudhury. I am a smartwatch and i can track my user's arm. In *Proceedings of the 14th Annual International Conference on Mobile Systems, Applications, and Services*, MobiSys '16, page 85–96, New York, NY, USA, 2016. Association for Computing Machinery. ISBN 9781450342698. doi: 10.1145/2906388.2906407. URL <https://doi.org/10.1145/2906388.2906407>.
- [6] Lilin Xu, Keyi Wang, Chaojie Gu, Xiuzhen Guo, Shibo He, and Jiming Chen. Gestureprint: Enabling user identification for mmwave-based gesture recognition systems, 2024. URL <https://arxiv.org/abs/2408.05358>.
- [7] Cheng Zhang, Yibin Hou, Jian He, and Xiaoyang Xie. Gesture recognition with focuses using hierarchical body part combination. *Tsinghua Science and Technology*, 30(4):1583–1599, 2025. doi: 10.26599/TST.2024.9010059. URL <https://www.sciopen.com/article/10.26599/TST.2024.9010059>.

APPENDIX

Project Code Repository

<https://github.com/kwang1012/CS598-Final-Project/tree/main>